Strength-two trades and transversal cubic gates: the Clebsch cubic-phase codes and their magic

Draft, September 2026

Abstract

A signed point configuration whose moments vanish through degree two defines a CSS code with a transversal cubic phase. We apply this dictionary to conic matchings, obtaining error-detecting codes with invariant-theoretic logical phases, and give an elementary $[[2p, p-1, 2]]_p$ construction for every prime $p \geq 5$. The Pauli spectrum of a homogeneous cubic phase is determined by its Hessian-rank distribution. Exact finite censuses distinguish the conic resources from products of elementary cubic states; the ten-qudit example over $\mathbb{F}_{11}$ is not Clifford-equivalent to a product across any bipartition. For the six-qudit example over $\mathbb{F}_7$, a native fourteen-input factory costs fewer than 16.116 raw resources per accepted block, compared with lower bounds of 54 and 36 for two explicitly specified separate-distillation menus at the same target block infidelity. A five-dimensional restriction identifies a chordal cubic and gives the two associated sign choices distinct operational meanings.

1 Signed trades as quantum codes

The same cancellation that makes two point configurations agree on quadratics can make a cubic phase insensitive to a stabilizer coordinate. The surviving third moment is then a logical gate. Our examples use perfect matchings of a conic, but the cancellation mechanism applies to any signed configuration.

Fix a prime $p \geq 5$ and write $\omega = \exp(2\pi i/p)$. A qudit has basis $\{|x\rangle : x \in \mathbb{F}_p\}$, with $X(a)|x\rangle = |x+a\rangle$ and $Z(b)|x\rangle = \omega^{bx}|x\rangle$. A *strength-two signed trade* is a list of distinct points $x_i \in \mathbb{F}_p^k$, with signs $\epsilon_i \in \{1, -1\}$, such that

$$\sum_i \epsilon_i = 0, \qquad \sum_i \epsilon_i x_i = 0, \qquad \sum_i \epsilon_i x_i x_i^T = 0. \tag{1}$$

These equalities are in $\mathbb{F}_p$. Let E have rows x_i^T and let $L = \{a\mathbf{1} + Eu : a \in \mathbb{F}_p,\ u \in \mathbb{F}_p^k\} \subseteq \mathbb{F}_p^n$. The coordinatewise product span is denoted by $L^{\circ 2}$.

Theorem 1.1 (Trade-to-code dictionary)**.** *Suppose $n = 2p$, $k = p-1$, and $\dim L = p$. Then the CSS code with X-stabilizer space $S = \langle \mathbf{1} \rangle$ and Z-stabilizer space $L^\perp$ has parameters $[[2p, p-1, 2]]_p$. The signed transversal operation $\bigotimes_i M_{\epsilon_i}$, where $M_c|z\rangle = \omega^{cz^3}|z\rangle$, induces the logical diagonal gate*

$$U_F|u\rangle = \omega^{F(u)}|u\rangle, \qquad F(u) = \sum_i \epsilon_i (x_i^T u)^3. \tag{2}$$

More generally, for $S \subseteq L$, a physical cubic with weights w_i is constant on every coset of S in L if and only if $\sum_i w_i s_i b_i c_i = 0$ for all $s \in S$ and $b, c \in L$. For $S = \langle \mathbf{1} \rangle$ this condition is $w \perp L^{\circ 2}$.

Proof. Put $D = \operatorname{diag}(\epsilon_i)$. Equation (1) says that L is isotropic for the nondegenerate pairing $v^T Dw$. Its dimension is half the length, so $L^\perp = DL$. In particular $S \subseteq L$, and the CSS dimension is $\dim L - \dim S = p - 1$. An encoded basis is

$$|u_L\rangle = p^{-1/2} \sum_{a \in \mathbb{F}_p} |a\mathbf{1} + Eu\rangle.$$

Expanding $\sum_i \epsilon_i (a + x_i^T u)^3$ removes all terms involving a by (1), leaving (2).

For the distance, no weight-one Z error lies in $S^\perp$. Since the points are distinct, $e_i - e_j \in S^\perp \setminus L^\perp$ for every $i \neq j$, so $d_Z = 2$. A weight-one vector cannot lie in L: its pairing with $\mathbf{1} \in L$ under D would be nonzero. Thus $d_X \geq 2$.

For the general criterion, write $T_w(s, b, c) = \sum_i w_i s_i b_i c_i$. The difference of the physical cubic at $z + s$ and z is $3T_w(s, z, z) + 3T_w(s, s, z) + T_w(s, s, s)$. Its quadratic part must vanish as a polynomial function on L; degrees are less than p. Polarization, using $2, 3 \neq 0$, gives $T_w(s, L, L) = 0$. This condition also kills the lower terms because $s \in L$, proving sufficiency. □

This is a specialization of the signed qudit orthogonality mechanism in Watson–Campbell–Anwar–Browne [1, Definition 2 and Lemmas 6–8], with related divisible-code and prime-qudit constructions in [2, 3, 4]. The contribution considered here is the choice of configuration and the resulting logical resource. Our positioning follows the bounded comparison recorded in ledger rows N1–N4 of the accompanying literature audit; no classification of all CSS codes with these parameters is asserted. The Schur-square conductor also governs the MDS–CSS transversal-group classification [22]. The distinction is the full product $S \circ L \circ L$ in Theorem 1.1: Schur-square codimension one alone does not imply a non-Clifford gate. Here the X-space is only $\langle \mathbf{1} \rangle$.

1.1 An elementary family at every prime

An example separates parameter existence from the conic geometry. Let $V = \mathbb{F}_p^p / \langle \mathbf{1} \rangle$, write a_i for the image of the ith standard basis vector, and set $t = a_0$. The functional $\sigma([z]) = \sum_i z_i$ is well-defined because $p = 0$ in $\mathbb{F}_p$. Take the positive sheet to be $\{a_i\}$ and the negative sheet to be $\{a_i + t\}$.

Proposition 1.2 (Translation trade)**.** *These sheets define a strength-two trade with affine span V and $\dim L = p$. Its code is $[[2p, p-1, 2]]_p$, and its logical cubic is*

$$F(u) = -3u(t)q(u), \qquad q(u) = \sum_i u(a_i)^2 \quad (u \in V^*). \tag{3}$$

The quadratic form q has rank $p - 2$. Moreover $\dim L^{\circ 2} = 2p - 1$; every preserving physical cubic weight is a scalar multiple of the sheet-sign vector.

Proof. The sheets lie in the disjoint affine hyperplanes $\sigma = 1, 2$. Within a sheet the points are distinct, and their differences span $\ker \sigma$; t supplies the remaining direction. Since $\sum_i a_i = 0$, translation preserves the first two moments of this p-point set. The cubic expansion gives (3). Identify V^* with $K = \{u \in \mathbb{F}_p^p : \sum_i u_i = 0\}$. Then $q(u) = \sum_i u_i^2$; its radical on K is $\langle \mathbf{1} \rangle$, so its rank is $p - 2$ and the displayed cubic is nonzero.

For rigidity, let weights α_i, β_i on the two sheets annihilate every affine quadratic. Put $\gamma = \alpha + \beta$ and $B = \sum_i \beta_i$. The linear moment says $\gamma + Be_0 \in \langle \mathbf{1} \rangle$. The quadratic moment on K has symmetric matrix

$$A = \operatorname{diag}(\gamma) + e_0 \beta^T + \beta e_0^T + B e_0 e_0^T.$$

A symmetric form vanishing on $K \times K$ is $\mathbf{1}r^T + r\mathbf{1}^T$ for some r. For distinct $i, j \neq 0$, comparison gives $r_i + r_j = 0$. There are at least four such indices, whence all $r_i = 0$ for $i \neq 0$. Diagonal comparison gives $\gamma_i = 0$ there; the linear moment then forces $\gamma_0 = -B$. Off-diagonal comparison gives $\beta_i = r_0$ for $i \neq 0$, and the $(0,0)$ entry gives $\beta_0 = r_0$. Thus $B = pr_0 = 0$, $\gamma = 0$, and $\alpha = -\beta$ are constant weights. The annihilator of $L^{\circ 2}$ is therefore exactly the sign line. $\square$

1.2 The conic configurations

For the two symmetric examples take $Q = XZ - Y^2$ and label its points by $a \mapsto (1 : a : a^2)$ and $\infty \mapsto (0 : 0 : 1)$. For a perfect matching M of $\mathbb{P}^1(\mathbb{F}_p)$ define

$$P_M = \prod_{\{a,b\} \in M} L_{ab}, \quad L_{ab} = abX - (a+b)Y + Z, \quad L_{a\infty} = aX - Y.$$

Use base matchings

$$M_7 = \{02, 14, 3\infty, 56\}, \qquad M_{11} = \{01, 25, 37, 49, 68, 10\infty\}.$$

Their $\mathrm{PGL}_2(p)$ orbits have $2p$ members. The determinant square class of a transformation taking the base matching to M specifies its sign. The coefficient vector of $(P_M - P_{M_p})/Q$ specifies x_M. The exact division, sign consistency, and moment conditions are part of the finite input verification. This is the configuration studied in *Quadratic Trade Rigidity and Cubic Orientation in Conic Matching Quotients* [20].

Proposition 1.3 (Finite conic input)**.** *For the stated matchings the construction has distinct points,* $\dim L = p$*, and* $\dim L^{\circ 2} = 2p - 1$*. It gives* $[[14, 6, 2]]_7$ *and* $[[22, 10, 2]]_{11}$*. In invertible logical coordinates the phases are*

$$F_7 = 4sI + 3J, \qquad F_{11} = 4sI_2 + 2I_3. \tag{4}$$

Here $I = ae - 4bd + 3c^2$ *and* $J = \det \begin{pmatrix} a & b & c \\ b & c & d \\ c & d & e \end{pmatrix}$ *are binary-quartic invariants. For the binary octavic* $f(S,T) = \sum_{i=0}^{8} \binom{8}{i} z_i S^{8-i} T^i$, $I_2 = (f,f)_8$ *and* $I_3 = ((f,f)_4, f)_8$ *use normalized transvectants.*

Finite verification. Appendix A specifies the matrices and substitutions. Exact finite-field expansion checks the moments, ranks and both polynomial identities. The signed-trade theorem then supplies the code and gate. These are finite polynomial and linear-algebra checks; the evidence record gives the independent reconstruction and its limits. $\square$

The invariant here is the logical phase itself. This differs from using invariant theory to study weight enumerators [5]; it is not a claim that invariant-defined gates in general originate with these examples.

2 The Hessian spectrum and product exclusion

For a homogeneous cubic F on $\mathbb{F}_p^k$, put $|F\rangle = p^{-k/2} \sum_u \omega^{F(u)} |u\rangle$ and $H_F(v) = (\partial_i \partial_j F)(v)$. We use natural logarithms and define

$$M_\alpha(|\psi\rangle) = \frac{1}{1-\alpha} \log \left(p^{-k} \sum_{v,w} |\langle \psi | X(v) Z(w) | \psi \rangle|^{2\alpha} \right) \quad (\alpha > 0,\ \alpha \neq 1).$$

These are stabilizer Rényi entropies with the normalization in [6, 7]; they vanish on pure stabilizer states and are additive under tensor products. The following quadratic-sum calculation makes them accessible without enumerating all p^{2k} Paulis.

Theorem 2.1 (Rank formula)**.** *Let $p \geq 5$, and set $r(v) = \operatorname{rank} H_F(v)$ and $N_r = \#\{v : r(v) = r\}$. Then*

$$|\langle F|X(v)Z(w)|F\rangle| = \begin{cases} p^{-r(v)/2}, & w \in \operatorname{im} H_F(v), \\ 0, & \textit{otherwise}. \end{cases} \tag{5}$$

Consequently

$$M_\alpha(|F\rangle) = \frac{1}{1-\alpha} \log \left(p^{-k} \sum_{r=0}^{k} N_r p^{(1-\alpha)r} \right). \tag{6}$$

Proof. The expectation is $p^{-k} \sum_u \omega^{F(u)-F(u+v)+w^T u}$. Writing $c_v = \nabla F(v)$, the exponent, up to a constant, is $-\frac{1}{2}u^T H_F(v)u + (w-c_v)^T u$. The sum over the radical vanishes unless $w - c_v \in \operatorname{im} H_F(v)$. Euler's identity $H_F(v)v = 2c_v$ makes this equivalent to the condition in (5). On a complementary subspace diagonalize the quadratic form. Each nonzero quadratic coefficient contributes a Gauss sum of modulus $\sqrt{p}$: its squared modulus follows by replacing a pair (x, y) with $(x-y, x+y)$ in the double sum, an invertible substitution. The radical contributes $p^{k-r(v)}$, proving the modulus formula. There are $p^{r(v)}$ admissible w for each v; summing their powers proves (6). □

Kagamihara–Tsuchiya [8, Theorem 1] give the binary rank-reduction formula, with $C(x) + C(x)^T$ replacing the odd-characteristic Hessian. Chen–Yan–Zhou [9] supply related hypergraph-state moment methods. We use the established rank-reduction method in odd characteristic; no first rank-formula or first odd-prime extension claim is made (ledger N5).

Proposition 2.2 (Exact conic spectra)**.** *The nonzero entries of the vector rank distributions for the two conic phases are*

$p = 7:\ r$	0	3	4	5	6
N_r	1	48	2940	26502	88158

$p = 11:\ r$	0	5	6	7	8	9	10
N_r	1	120	14520	80520	21442960	2387485210	23528401270.

In each case the minimum positive rank is $(p-1)/2$, attained on exactly $p+1$ projective points. In the binary-form coordinates these are $s = 0$ and the perfect $(p-3)$rd powers. In particular,

$$M_2(F_7) = \log \frac{1977326743}{78835}, \qquad M_2(F_{11}) = \log \frac{11^{19}}{7151076991}.$$

Finite verification and deduction. The census visits vectors with first nonzero coordinate one. Since $H_F(av) = aH_F(v)$ for $a \neq 0$, every projective representative contributes $p-1$ vectors; the zero vector is added separately. The domains have 19608 and 2593742460 representatives. The $p = 7$ census has an independent implementation; the $p = 11$ exhaustive execution is checked against it in the smaller case, not independently duplicated at its full size. Substitution verifies the $p+1$ perfect-power points. The census has exactly that many minimum-rank lines, so this containment is an equality over $\mathbb{F}_p$. Equation (6) gives the entropy values. □

Lemma 2.3 (Product ceiling)**.** *For a pure state in dimension D with an orthogonal unitary Pauli basis, $M_2 \leq \log((D+1)/2)$. Thus a pure product on blocks of a and $k-a$ prime-dimensional qudits satisfies*

$$M_2 \leq \log \frac{(p^a+1)(p^{k-a}+1)}{4}. \tag{7}$$

Proof. Put $x_P = |\langle P\rangle|^2$. Purity gives $\sum_P x_P = D$ and $x_I = 1$. Cauchy–Schwarz on the D^2-1 other entries yields $\sum_P x_P^2 \geq 1 + (D-1)^2/(D^2-1) = 2D/(D+1)$. Insert this into the definition of M_2. For a product the Pauli sum factorizes, so the entropies add. $\square$

The product-ceiling method has predecessors in the characteristic-function and qudit entropy literature [10, 7]. The argument above needs no assumption that a SIC fiducial exists or attains the bound.

Corollary 2.4 (Clifford-product exclusion)**.** *The ten-qudit state $|F_{11}\rangle$ is not Clifford-equivalent to a product across any nontrivial bipartition. The same conclusion holds for the dihedral and Borel six-qudit conic trades over $\mathbb{F}_7$ in Appendix A. This entropy test alone does not establish the conclusion for the six-qudit Clebsch state $|F_7\rangle$.*

Proof. For fixed k, the largest right side of (7) occurs at $a = 1$ or $k-1$: the variable part $p^a + p^{k-a}$ increases toward the endpoints. At $p = 11, k = 10$ the bound is $22.6797\ldots$, below $22.8695\ldots = M_2(F_{11})$. At $p = 7, k = 6$ it is $10.4228\ldots$; the other two trades give $10.4246\ldots$ and $10.4840\ldots$, whereas $M_2(F_7) = 10.1299\ldots$. These comparisons are checked as exact rational inequalities before taking logarithms. Clifford conjugation permutes the Pauli moduli, so it preserves M_2. $\square$

For comparison, independent cubic qudits have $M_2 = k\log(p^2/(2p-1))$. Two disjoint xyz phases at $p = 7$ have $M_2 = 8.8047\ldots$, also below F_7. The trace cubic $F(u) = \mathrm{Tr}_{\mathbb{F}_{p^k}/\mathbb{F}_p}(u^3)$ has full Hessian rank at every nonzero u, by nondegeneracy of the trace pairing, and hence $M_2 = \log(D^2/(2D-1))$, $D = p^k$. These finite-field cubic fiducials belong to the Alltop–MUB construction [11, 12, 13]. Feng–Luo [14, Proposition 2 and Appendix C] prove single-prime L1 optimality among diagonal gates. Neither that result nor the L1 ceiling of [13] is an unrestricted M_2 maximality theorem. The conjecture in [15] concerns two prime-dimensional qudits; we do not extend its attribution to arbitrary k (ledger N9).

3 A bounded factory comparison

The resource-count question is different from the entropy question. We compare preparing and purifying the whole six-qudit phase with purifying independent single-qudit primitives and then compiling it. This is the synthillation strategy of Campbell–Howard [16]; Li–Yeh [17, Section 4.4 and Appendix C.1] already connects signed physical qutrit gates to a coupled logical resource and its distillation. Our claim is the following finite, same-target comparison (ledger N8).

3.1 Noise and allowed operations

Every raw input is

$$\rho_c = (1-\delta)|M_c\rangle\langle M_c| + \frac{\delta}{6}\sum_{e\neq 0} Z(e)|M_c\rangle\langle M_c|Z(-e), \qquad |M_c\rangle = 7^{-1/2}\sum_x \omega^{cx^3}|x\rangle. \tag{8}$$

Inputs are independent; stabilizer preparation, Clifford operations, measurements, feed-forward and storage are ideal. Uniformity of the six nonzero labels is an assumption, not a consequence of diagonal twirling.

The native factory prepares $|+_L\rangle = 7^{-7/2}\sum_{z\in L}|z\rangle$, injects the fourteen signed cubic gates, measures the code stabilizers, and decodes on acceptance. Injection is deterministic: controlled subtraction from data x to a magic ancilla, followed by ancilla outcome m, supplies phase $c(x+m)^3$.

A quadratic Clifford correction removes the m-dependent terms. An ancilla error $Z(e)$ becomes $Z(e)$ on the data. No injection outcome is discarded. For the resulting physical error vector e, acceptance is $\mathbf{1}^T e = 0$ and the logical error label is $E^T e$; the error is harmless exactly when $e \in L^{\perp}$.

The separate architecture distills each primitive independently with any finite concatenation of the three modules in Table 1, then uses deterministic linear-form cubic synthesis without final postselection. Outputs may be stored separately and successful attempts retained. Module choices and depths may differ between terms. A final Clifford is allowed if its input is a homogeneous cubic-phase state on the same six qudits. Ancilla-assisted general synthesis, pooled multi-output distillers and joint synthillation are outside this menu.

Module	points	X space	logical row	cubic weights
Weighted four-site	$0, 1, 2, 3$	$\langle 1 \rangle$	x	$(-1, 3, -3, 1)$
QRM six-site	$\mathbb{F}_7^*$	$\langle x \rangle$	1	all 1
RS seven-site	$\mathbb{F}_7$	$\langle 1, x \rangle$	x^2	all 1

Table 1: The three allowed one-output distillers. In each row $L = S + \langle \ell \rangle$ and the logical coefficient is -1; coordinate negation supplies $+1$. Parameters are respectively $[[4,1,2]]_7$, $[[6,1,2]]_7$, and $[[7,1,3]]_7$.

The six-site row is the prime-qudit QRM construction [3, 19]; the length-seven RS row is checked separately. We distinguish raw supplies. In the *sign-only menu*, only $c = \pm 1$ has unit cost. The four-site module costs eight raw injections because each ± 3 gate uses three signs. Its two middle error rates are $\delta_3 = (6/7)[1-(1-7\delta/6)^3]$. In the *weighted relaxation*, every nonzero cubic coefficient has unit raw cost and error δ. This relaxation favors the separate architecture.

Lemma 3.1 (Cubic synthesis lower bound)**.** *The weighted linear-form cubic ranks obey* $9 \le r(F_7) \le 13$ *and* $15 \le r(F_{11}) \le 21$*. The rank-nine lower bound also holds for any homogeneous cubic phase on six qudits Clifford-equivalent to* $|F_7\rangle$*.*

Proof. In a shortest decomposition $F = \sum_{i=1}^{r} c_i \ell_i^3$, the coefficient vectors span the logical space and are projectively distinct. The Hessian at v has rank at most the number of $\ell_i(v) \neq 0$. For F_7, killing five independent forms gives $r \ge 8$. If $r = 8$, every six coefficient vectors must be independent, or a common annihilator would have Hessian rank at most two. The $\binom{8}{5} = 56$ five-subsets then span distinct hyperplanes, producing 56 minimum-rank lines. The census has only eight. Thus $r \ge 9$. For F_{11} the same argument gives $r \ge 14$ and excludes equality: $\binom{14}{9} = 2002 > 12$ minimum-rank lines would be needed. The raw signed decompositions have one zero row, giving the upper bounds. For a cubic phase the Pauli-modulus multiplicity at $p^{-r/2}$ is $N_r p^r$, so the histogram recovers the rank census. Clifford invariance therefore preserves every input to the lower-bound argument. □

A weighted cube is implemented by Clifford computation of ℓ_i, one weighted cubic gate, and uncomputation. This signature-tensor synthesis framework is explicit in Heyfron–Campbell [18, Section IV]; the rank bound is for that model, not arbitrary circuits (ledger N7). For these matching configurations, the signed cubic line is also the Macaulay inverse system of the Artinian reduction with Hilbert function $(1, p-1, p-1, 1)$ [20, Corollary on self-association and cubic duality]. Thus the same cubic enters the geometric duality and the linear-form compilation problem.

Theorem 3.2 (Same-target resource comparison)**.** *Under* (8)*, for every* $0 < \delta \le 0.01$*, let* q *be the native factory's output block infidelity. Its expected raw consumption is less than* 16.116*. Every separate construction in the sign-only menu with block infidelity at most* q *costs at least* 54 *raw* *inputs per output; in the weighted relaxation it costs at least* 36. *Feasible separate constructions meeting this target exist throughout the interval.*

Proof. Write $a = 1 - \delta$ and P for native acceptance. Single errors are rejected. Every accepted weight-two error is harmful, since the evaluation points are distinct. Their probability is $(91/6)\delta^2 a^{12}$. For a fixed error support of size at least two, fixing all but its last nonzero label leaves at most one of six choices that accepts. A union bound on pairs therefore gives

$$a^{14} \le P \le 1 - 14\delta a^{13}, \qquad \frac{91\delta^2 a^{12}}{6P} \le q \le \frac{91\delta^2}{6a^{14}}. \tag{9}$$

Bernoulli's inequality implies

$$a^{12}(1 + 14\delta a) \ge (1 - 12\delta)(1 + 14\delta - 14\delta^2) = 1 + \delta(2 - 182\delta + 168\delta^2) \ge 1.$$

Hence $a^{12} \ge 1 - 14\delta a^{13} \ge P$, so $q \ge (91/6)\delta^2$, while $q/\delta \le (91/600)/(0.99)^{14} < 1$. Also $14/P \le 14/(0.99)^{14} < 16.116$.

A noisy primitive on a nonzero form with coefficient vector b contributes an independent error label eb. For probability distributions on the additive logical-label group, $\|\mu * \nu\|_\infty \le \|\mu\|_\infty$, since each convolution value is an average. An unpurified raw term therefore forces final error at least δ, even with cancellations or a fixed Pauli correction. The states $Z(\eta)|F\rangle$ are orthogonal, so this probability is exactly a block-infidelity statement. Since $q < \delta$, all synthesis terms must be purified. Each independent tree has at least six raw leaves in the sign-only menu and four in the weighted menu. Retrying cannot lower those counts. Lemma 3.1 requires at least nine terms, giving 54 and 36.

For feasibility use the thirteen-term signed decomposition. A weighted four-site primitive has error at most δ^2/a^4, so thirteen give block error at most $13\delta^2/(0.99)^4 < (91/6)\delta^2$. For the seven-site module harmful accepted errors have weight at least three, giving error at most $35\delta^3/a^7$; thirteen give $455\delta^3/a^7 < (91/6)\delta^2$ throughout the interval. A union bound suffices for these feasible upper estimates; the lower bound already allowed cancellations. $\square$

At $\delta = 0.01$ the exact enumerator calculation gives the comparison below. The separate rows are feasible constructions, not claims to attain their menu's optimum.

Construction	block infidelity	expected raw inputs
Native fourteen-input	0.00159853	16.0894
Weighted four-site, thirteen terms	0.00133141	54.1275
Sign-only seven-site, thirteen terms	0.0000130920	97.6325

For an ordinary enumerator A_w of $L^\perp$, native good-output probability before conditioning is $\sum_w A_w a^{14-w}(\delta/6)^w$. Acceptance also has the closed form $P = [1 + 6(1 - 7\delta/6)^{14}]/7$. Thus the calculation includes harmless accepted errors, rather than treating every nonzero physical error as failure.

4 A chordal restriction of the logical cubic

The logical cubic also provides an interface with the invariant pencil in *Chordal and Conference Cubics: Reconstruction and a Residual C_2-Torsor* [21]. A restriction of a logical polynomial is a resource on fewer variables; it does not by itself construct a subcode.

Proposition 4.1 (Chordal shadow and sign choices). *Over $\mathbb{F}_{11}$ the stabilizer A_5 of the base matching acts on the logical space as $\mathbf{1}\oplus V_4\oplus V_5$. The restriction to its unique five-dimensional constituent is, in the recorded coordinates,*

$$8\det\begin{pmatrix} z_0 & z_1 & z_2 \\ z_1 & z_2 & z_3 \\ z_2 & z_3 & z_4 \end{pmatrix}.$$

It is a chordal member of the invariant pencil. At ranks $0, 3, 4, 5$, the chordal and conference censuses are respectively

$$(1, 120, 27720, 133210), \qquad (1, 300, 22260, 138490).$$

Thus their five-qudit phase states are not Clifford-equivalent. At $p = 7$ the restriction $s = 0$ in (4) is $3J$. Reversing the sheets sends U_F to U_F^{-1}. The residual involution exchanging the two chordal members acts as a linear Clifford change of frame on the five shadow variables.

Finite identification and operational deduction. Character idempotents in the reconstructed A_5 action have ranks $1, 4, 5$. The recorded intertwiner identifies V_5 with the augmentation module on the six matching pairs; polynomial substitution gives the Hankel determinant. The two exact censuses then separate the resources by Theorem 2.1. The $p = 7$ restriction follows directly from (4). Sheet reversal changes every sign in (2), hence negates F. For an invertible linear map q, the basis permutation $|z\rangle \mapsto |qz\rangle$ is Clifford: it sends $X(v)$ to $X(qv)$ and $Z(w)$ to $Z(q^{-T}w)$. The recorded involution exchanges the two chordal polynomials, giving the stated frame change. The identity is on the shadow, not a claim about all extensions to the full ten-qudit gate. □

The singular scheme of the Hankel cubic is a rational normal quartic; its twelve $\mathbb{F}_{11}$-points are not the entire scheme. The existing lift tests rule out the identity-on-complement extension and geometric $\mathrm{PGL}_2(11)$ lifts of the residual involution. They do not classify arbitrary full-gate lifts. These boundaries are part of ledger N4.

Questions suggested by the construction

The translation family shows that the parameters and cubic-weight rigidity alone do not select the symmetric examples. Their distinction lies in the phase geometry and its spectrum. It remains to explain the observed stopping of the conic translation-class source beyond $p = 13$, rather than merely extend a matching enumeration. A second question is whether a balanced $2p$-point trade can realize a trace-cubic resource on $p-1$ logical qudits. Finally, the distance-three search inside the fixed $p = 7$ evaluation space fails, while the elementary distance-three RS examples do not establish the same product exclusion. Finding a distance-at-least-three code with several logical qudits and such an invariant phase is a separate construction problem. The present factory comparison leaves pooled distillation and unrestricted adaptive preparation open.

A Finite data and verification boundary

The arguments proving the signed-moment dictionary, translation family, Hessian formula, product ceiling and interval comparison are given in the text. The conic matrices, polynomial identities, exact spectra and shadow identification use finite computations. No Lean coverage is claimed for this companion. The source annotations identify each dependence; the claim map pins every theorem statement and records formal coverage as absent.

A.1 Coordinates and exact inputs

The evidence index supplies the full matrices, signs and raw cubics from the committed reconstruction bundle. For $p = 7$ the coefficient order is $X^2, XY, XZ, Y^2, YZ, Z^2$ and the logical substitution is $u = (a, b, s + c, 3s + c, d, e)$. For $p = 11$ the selected coefficient order is

$$X^4, X^3Y, X^3Z, X^2YZ, X^2Z^2, XY^2Z, XYZ^2, XZ^3, YZ^3, Z^4;$$

the substitution is $u = (z_0, z_1, 7z_2, 6z_3, 3s + 6z_4, s + 3z_4, 6z_5, 7z_6, z_7, z_8)$. The transvectant normalization is

$$(f, g)_r = \frac{(m-r)!(n-r)!}{m!n!} \sum_{j=0}^{r} (-1)^j \binom{r}{j} \frac{\partial^r f}{\partial S^{r-j} \partial T^j} \frac{\partial^r g}{\partial S^j \partial T^{r-j}},$$

for binary forms of degrees m, n; all denominators used here are invertible.

For the other $p = 7$ trades, one may reconstruct the positive and negative sheets as translation classes of the following matchings. Entries are partner lists on $0, \ldots, 6, \infty$, with index 7 representing infinity:

	$+$	$-$
dihedral	$(1, 0, 5, 4, 3, 2, 7, 6)$	$(1, 0, 6, 4, 3, 7, 2, 5)$
Borel	$(1, 0, 4, 6, 2, 7, 3, 5)$	$(1, 0, 6, 5, 7, 3, 2, 4)$.

Their vector censuses at ranks $0, 3, 4, 5, 6$ are respectively

$$(1, 6, 1092, 20202, 96348), \qquad (1, 6, 588, 20286, 96768).$$

Their fourth-moment sums are $58711/16807$ and $55327/16807$.

A.2 Classification and distance computations

The conic search enumerates perfect matchings, identifies their translation classes, and buckets classes by equal first and second moment tensors. Pairs in a bucket are tested for a nonzero third-moment difference, then quotiented by the stated affine action. The respective numbers of translation classes for $p = 5, 7, 11, 13, 17, 19$ are $3, 15, 945, 10395, 2027025, 34459425$; the numbers of inequivalent hits under $\mathrm{AGL}(1, p)$ are $0, 3, 2, 1, 0, 0$. This is a finite source classification, not a classification of all trades or all Clifford-equivalent codes. Every hit has the signed moment condition, $\dim L = p$ and Schur-square dimension $2p - 1$. The $p = 13$ hit gives $[[26, 12, 2]]_{13}$; its stored 20000-vector Hessian sample is not an exhaustive spectrum or a proved minimum-rank bound. The new $p = 11$ and $p = 13$ X-distance estimates are only upper bounds.

The fixed-space distance search uses all $L' \subseteq L$ containing $\mathbf{1}$ with $\dim L' \geq 3$. Put $R(L') = \{s \in L' : T(s, L', L') = 0\}$. It suffices to test the maximal admissible X-space $R(L')$. A weight-two $e_i - e_j$ is undetected exactly when columns i, j of this space agree; it is harmless exactly when those columns of L' agree. Thus distance at least three requires equality of the two column partitions. Reduced row-echelon enumeration of subspaces of $L/\langle \mathbf{1} \rangle$ visits 61927311 candidates of dimensions two through six. None with a nonzero restricted cubic passes the partition test. The one-dimensional case also fails: a nonzero one-variable cubic has zero trilinear radical on that quotient, so its only available X-space is $\langle \mathbf{1} \rangle$, which does not separate its distinct evaluation columns. This excludes the specified nonzero-cubic subcodes inside the fixed L; it does not exclude other evaluation spaces, concatenation or code switching. For comparison, evaluation on all of $\mathbb{F}_p$ with X-space RS_2 gives the length-p distance-three rows $[[7, 1, 3]]_7$, $[[11, 3, 3]]_{11}$ and $[[13, 4, 3]]_{13}$ in the recorded polynomial-code search. They are distinct from Campbell's length-$(p-1)$ construction [19]; their checks belong to the strengthening bundle. The exhaustive negative has an internal count check and recorded tally, but no independent full second implementation.

A.3 Artifacts and reproducibility

The machine-readable evidence registry in `verification/evidence.json` records the exact repository paths, checksums, byte counts, replay commands, search domains and limits. The inputs are the committed C1090 reconstruction and spectrum bundles, the C1099 strengthening and shadow bundles, and the C1102 factory benchmark. The factory certificate contains all native and primitive enumerators, exact rational sample values and interval constants. Its checks compare primal enumeration with MacWilliams transforms and synthesis-kernel enumeration with the Fourier expression; no random seed or floating-point decision enters that benchmark.

The lightweight draft check verifies statement identities, annotation links, input hashes, census arithmetic and the exact product and interval inequalities. A fresh standard-library check also re-expands both coordinate normal forms, checks both conic moment and Schur-square identities, reproduces the complete $p = 7$ census, and checks seven primes in the translation family. It does not rerun the multi-billion-point $p = 11$ census or the matching exhaustions. Those remain explicitly attributed to the recorded executions; full commands and their runtime estimates accompany the evidence registry. This working draft resolves artifacts within the monorepo. A standalone archival package, public immutable locator and external manuscript review remain release work, and are not implied by the local draft check.

A.4 Literature coverage

The literature record has 336 explicit dispositions: 23 primary-comparison memberships, 47 abstract/metadata comparisons and 266 title screens. Its 33 technical sources include three full-text readings; the remaining read depths and exact sections are recorded individually. Title screening does not technically exclude the unread bodies. The canonical positioning ledger N1–N9 is retained in the citation audit, and all priority wording here is scoped to that ledger.

Seven original citation seeds have independent OpenAlex, Crossref and Semantic Scholar counts. For the added Kagamihara–Tsuchiya preprint, OpenAlex and Semantic Scholar return zero with empty enumerations; Crossref has no record for the DataCite-registered arXiv DOI. The explicitly approved coverage exception preserves that result as unavailable, with no three-source absence verdict. MathSciNet is not covered. This limits priority claims; it does not alter the positive mathematical attribution to the preprint.

References

[1] Qudit Colour Codes and Gauge Colour Codes in All Spatial Dimensions. arXiv:1503.08800.

[2] Towers of generalized divisible quantum codes. arXiv:1709.08658.

[3] Magic state distillation in all prime dimensions using quantum Reed-Muller codes. arXiv:1205.3104.

[4] Towards low overhead magic state distillation. arXiv:1811.08461.

[5] Invariant Theory, Magic State Distillation, and Bounds on Classical Codes. arXiv:2501.10163.

[6] Stabilizer Rényi Entropy. arXiv:2106.12587.

[7] Stabilizer Rényi entropy on qudits. 10.1007/s11128-023-04186-9.

[8] Stabilizer Rényi entropy of 3-uniform hypergraph states. arXiv:2602.23687.

[9] Magic of quantum hypergraph states. arXiv:2308.01886.

[10] Detecting Magic States via Characteristic Functions. 10.1007/s10773-022-05027-8.

[11] Complex sequences with low periodic correlations. 10.1109/TIT.1980.1056185.

[12] Constructions of Mutually Unbiased Bases. arXiv:quant-ph/0309120.

[13] Product Weyl-Heisenberg covariant MUBs and Maximizers of Magick. arXiv:2603.15550.

[14] Optimality of the Howard-Vala T-gate in stabilizer quantum computation. 10.1088/1402-4896/ad80e7.

[15] Analytical Landscape of Maximal Magic for Two-Qutrit States and Beyond. arXiv:2607.07197.

[16] A unified framework for magic state distillation and multi-qubit gate synthesis with reduced resource cost. arXiv:1606.01904.

[17] Transversal AND in Quantum Codes. arXiv:2603.04548.

[18] A quantum compiler for qudits of prime dimension greater than 3. arXiv:1902.05634.

[19] Enhanced fault-tolerant quantum computing in d-level systems. arXiv:1406.3055.

[20] Quadratic Trade Rigidity and Cubic Orientation in Conic Matching Quotients. Companion manuscript, repository source `papers/clebsch-factorization/`.

[21] Chordal and Conference Cubics: Reconstruction and a Residual C_2-Torsor. Companion manuscript, repository source `papers/chordal-conference-reconstruction/`.

[22] Diagonal Isoduality and Transversal Clifford Groups of MDS–CSS Codes. Companion manuscript, repository source `papers/mds_css_transversal_groups/`.